



“An LLM hallucinates because it does not have sufficient knowledge, i.e., not trained on the task that it was asked to do.”

— **Divya Venkatraman,**  
**Founder & CTO, DeepDive Labs**

examples of generative AI models providing slanderous information about celebrities, inexistent bank transactions, imaginary patient records, and incorrect company policies.

Such cases where AI models present factually incorrect, illogical, or fabricated responses—often in a very confident tone—are known as AI hallucinations.

Hallucinations are not new to AI—they have existed since before LLMs became popular. But now, we are more aware of them because the AI user base is growing, and more cases are in the limelight. “We tend to talk about hallucinations after the emergence of LLMs. It is important to note that LLMs are not the only form of AI models. LLMs are models that approximate language, i.e., they are trained on a large corpus of text to generate text in human-readable form with the right grammar structure. This means to users that LLMs are trained to generate text that ‘reads well’ as a response to one’s query. They are not trained to provide factually correct answers! British statistician George Box summarises this in his quote—All models are wrong, but some are

useful,” remarks Divya Venkatraman, founder & CTO of DeepDive Labs, a Singapore-based tech startup providing bespoke data and AI services in the APAC region.

With the rise in popularity of LLMs, they are being used for various use cases—from writing stories and generating visualisations to answering questions on various topics. Venkatraman explains that since LLMs fundamentally approximate language, they generate text that reads well but might not be accurate for all use cases. This phenomenon is known as the hallucination of language models. Traditional AI models also experienced hallucinations but these were referred to as ‘errors’ and were quantified using different metrics.

## Why do AI models hallucinate?

Most AI models tend to hallucinate because generative AI is inherently probabilistic and less semantic. These models are trained on data from the internet, which contains factual information and significant misinformation. Additionally, many AI models have knowledge cut-offs or lack proper grounding during response generation. Here is a quick run-through of the potential causes of AI hallucinations:

**Quality of training data:** The well-known thumb rule—garbage in, garbage out (GIGO)—applies here. If an AI model is trained on biased, incorrect, outdated, or otherwise faulty data, its responses will reflect those same issues.

**Extent of training:** Overtraining (or overfitting) and undertraining (or underfitting) can negatively impact an AI model. Overtrained models tend to memorise the training data rather than generalise patterns. When such a model encounters a different situation, it behaves like a student tackling off-syllabus questions, often hallucinating. Conversely, models that are too simple or trained

with limited data leave significant information gaps. These gaps are filled with statistically plausible but factually incorrect patterns. “An LLM hallucinates because it does not have sufficient knowledge, i.e., not trained on the task it was asked to do. LLMs tend to hallucinate in domain-specific questions pertaining to fields like law or medicine, for example,” explains Venkatraman.

## Model complexity and architectural limitations:

Neural network constraints and probabilistic architectures can lead to hallucinations, and so can poor design and bad programming!

Take sequential token generation, for example. LLMs generate text token by token, much like auto-complete. A single error propagates through subsequent tokens, compounding the mistake. If it makes one mistake, it has no way of going back and correcting it, so it keeps filling token after token, building on the erroneous one quite confidently—and results in a nonsensical response.

With increasing usage, experts are discovering newer points of failure like this and working to sort them out—and it is not easy. Developing a proper AI model requires deep knowledge of mathematical models and sound programming. Poor programming can hinder a model’s ability to discern patterns, while overly complex systems may struggle with decision-making. Both scenarios increase the risk of hallucination when facing insufficient data or unfamiliar contexts.

If someone tells you that developing an AI chatbot is now as easy as drag-and-drop, do not believe them. Your system will only be as good as the foundation.

**Bad prompts and poor understanding of context:** In an article in the Times of India, B Ravindran, Professor and Head of The Wadhwani School of Data Science and AI, IIT